

# Report

Title: Provisioning Elastic IP and managing association with an EC2 Instance

## Theory

### Elastic IP

- Static Public IP
- Reassign able
- High availability (minimum downtime)
- AWS charges for unused IP but if you assigned it to a machine, it is free of cost

## Setup

Create a fresh EC2 instance with following configs;

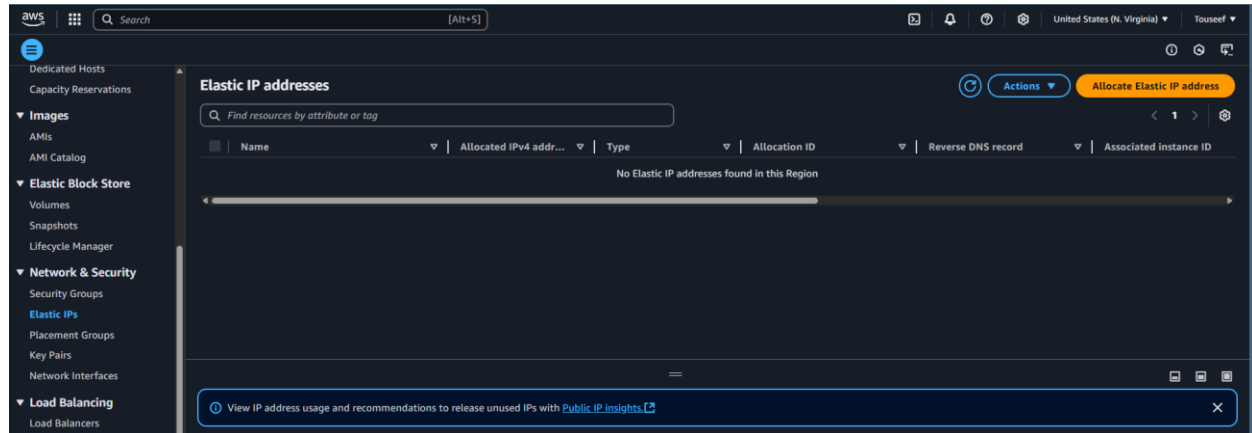
- AMI => Amazon Linux
- Instance Type => T3.micro
- No Key Pair needed
- Proceed with all Defaults

## Procedure

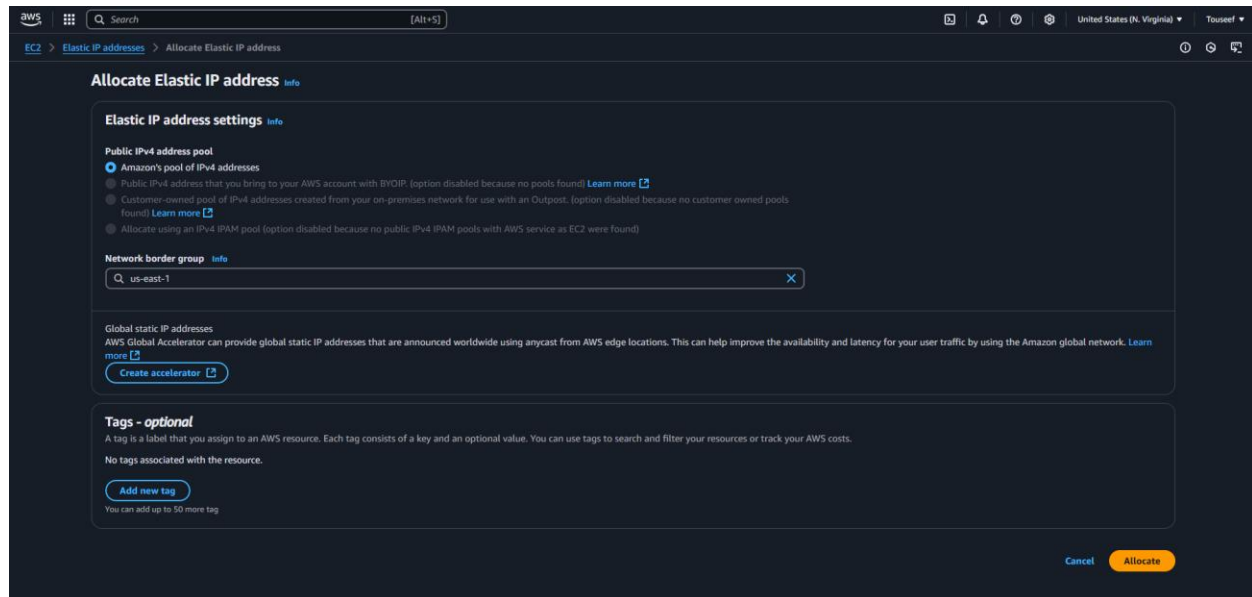
After creating an EC2 instance, note down the public IP;

The screenshot shows the AWS Management Console interface for an EC2 instance. At the top, there's a search bar and navigation tabs for 'Instances'. Below this, a table lists the instance 'elstic-ip-lab' with its ID 'i-0e8bf6bdfbe14af0', state 'Running', type 't2.micro', and public IPv4 address '44.211.40.130'. The instance is located in the 'us-east-1a' availability zone. Below the table, the 'Details' tab for the instance is expanded, showing various attributes like 'Instance ID', 'IPv6 address', 'Hostname type', 'Answer private resource DNS name', 'Auto-assigned IP address', 'Public IPv4 address', 'Instance state', 'Private IP DNS name (IPv4 only)', 'Instance type', 'VPC ID', 'Private IPv4 addresses', 'Public IPv4 DNS', 'Elastic IP addresses', and 'AWS Compute Optimizer finding'.

Now go to the Elastic Ips option in the sidebar menu. Here press the Allocate Elastic IP address button.

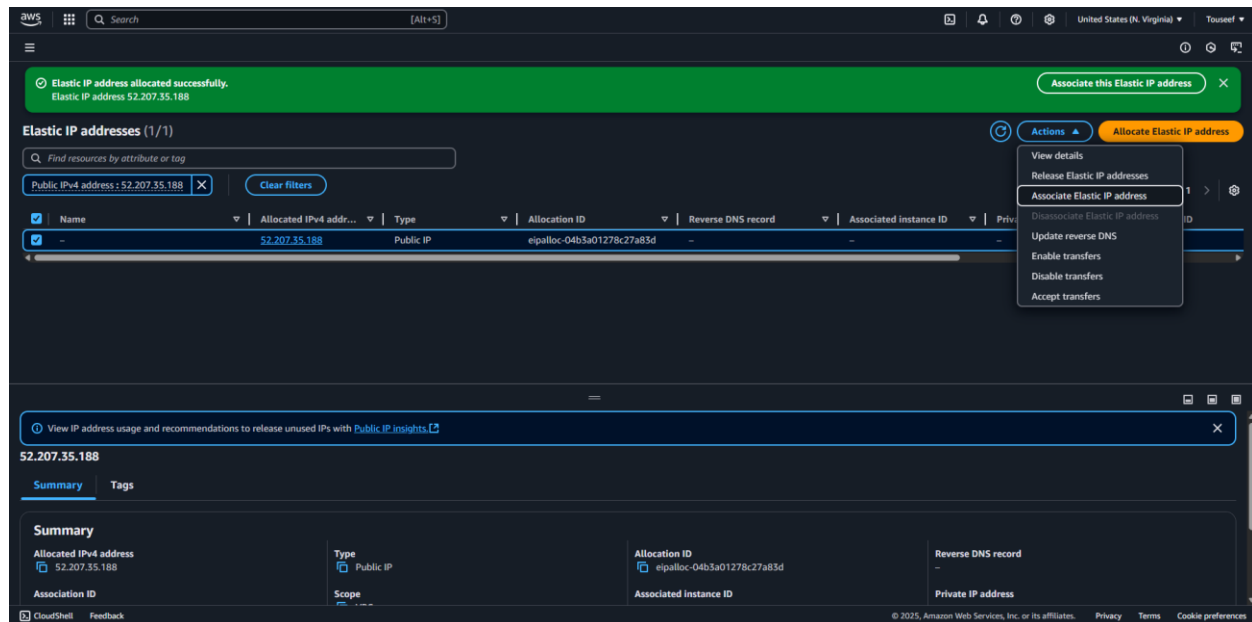


Proceed with default values and press Allocate.



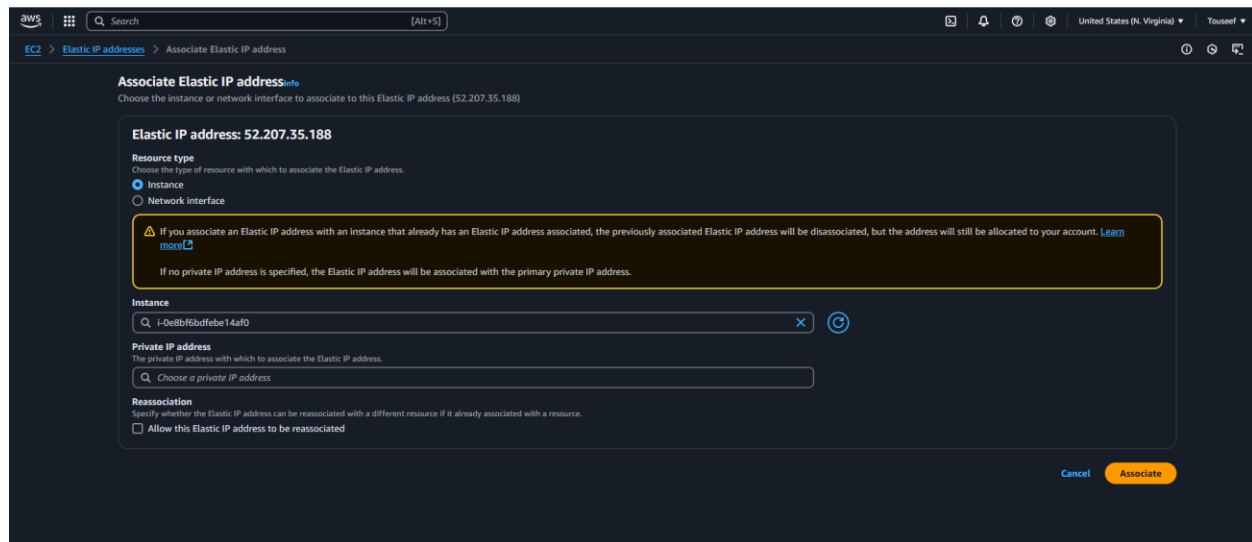
A random IPv4 public IP will be reserved in your AWS account as shown below. Now we need to assign it to our EC2 instance.

Select the Elastic IP that you want to associate, go to Actions, select Associate Elastic IP address option.



The screenshot shows the AWS Elastic IP addresses console. At the top, a green notification bar states "Elastic IP address allocated successfully. Elastic IP address 52.207.35.188" with a button to "Associate this Elastic IP address". Below this, the "Elastic IP addresses (1/1)" section displays a table with one entry: 52.207.35.188, Public IP, Allocation ID: eipalloc-04b3a01278c27a83d. The "Actions" menu is open, showing options: View details, Release Elastic IP addresses, Associate Elastic IP address (highlighted), Disassociate Elastic IP address, Update reverse DNS, Enable transfers, Disable transfers, and Accept transfers. Below the table, a "Summary" tab is selected, showing details for the Elastic IP address 52.207.35.188, including its Allocation ID and associated Instance ID.

Here select your EC2 instance and press Associate.



The screenshot shows the "Associate Elastic IP address" wizard in the AWS console. The Elastic IP address 52.207.35.188 is selected. Under "Resource type", "Instance" is chosen. A warning box states: "If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account." Below this, the "Instance" field is populated with "i-0e8bfbdfbe14a10". The "Private IP address" field is empty, with a note: "If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address." The "Reassociation" section has a checkbox "Allow this Elastic IP address to be reassociated" which is unchecked. At the bottom right, there are "Cancel" and "Associate" buttons.

Now go back to your EC2 instance details. Here you would see that the Public IP of the instance has been replaced with the Elastic IP.

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and user information. Below this, the 'Instances' page is displayed. A table lists the instances, with 'elastic-ip-lab' selected. The instance details panel on the right shows the following information:

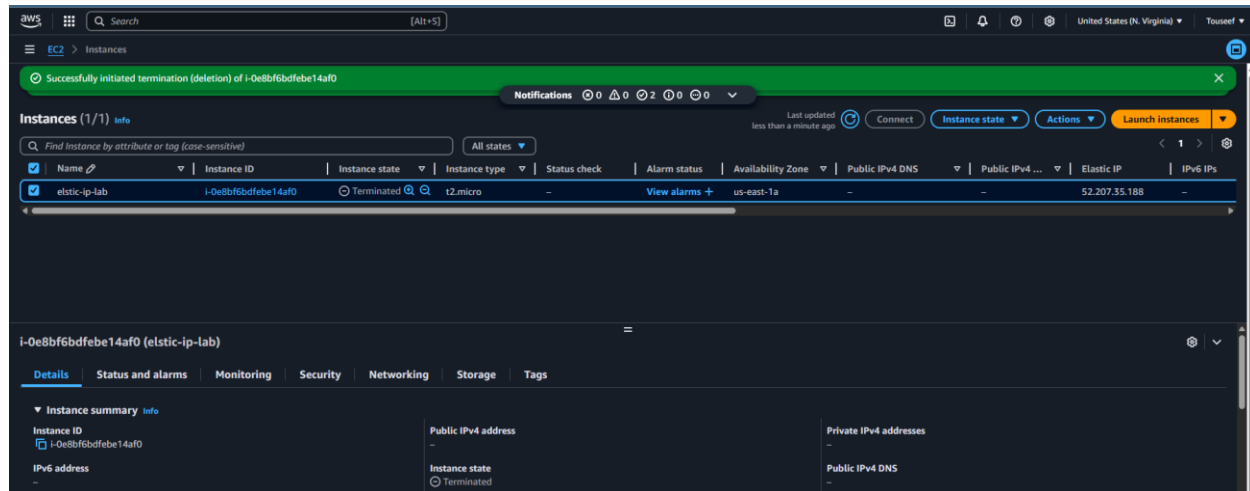
- Instance ID:** i-0e8bf6bdfbe14af0
- Instance state:** Running
- Instance type:** t2.micro
- Public IPv4 address:** 52.207.35.188
- Private IPv4 addresses:** 172.31.87.112
- Public IPv4 DNS:** ec2-52-207-35-188.compute-1.amazonaws.com
- Private IP DNS name (IPv4 only):** ip-172-31-87-112.ec2.internal
- Elastic IP addresses:** -

Now stop the instance. You would notice that the Elastic IP is still assigned to the instance.

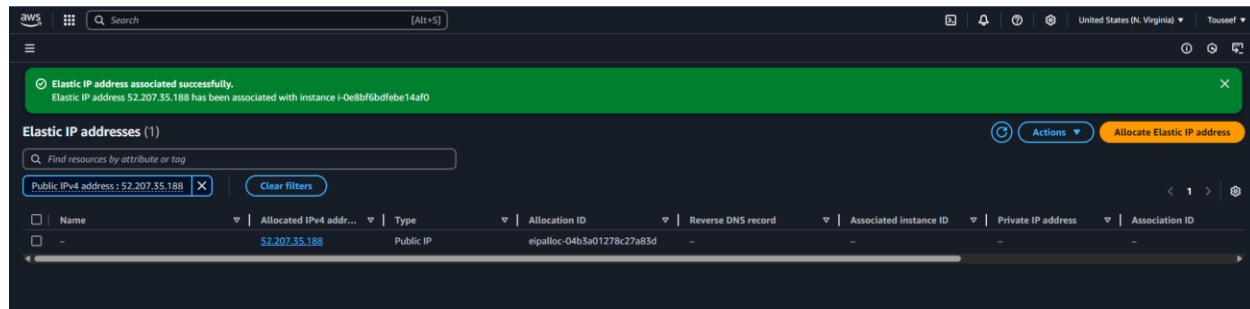
The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and user information. Below this, the 'Instances' page is displayed. A table lists the instances, with 'elastic-ip-lab' selected. The instance details panel on the right shows the following information:

- Instance ID:** i-0e8bf6bdfbe14af0
- Instance state:** Stopped
- Instance type:** t2.micro
- Public IPv4 address:** 52.207.35.188
- Private IPv4 addresses:** 172.31.87.112
- Public IPv4 DNS:** ec2-52-207-35-188.compute-1.amazonaws.com
- Elastic IP addresses:** -

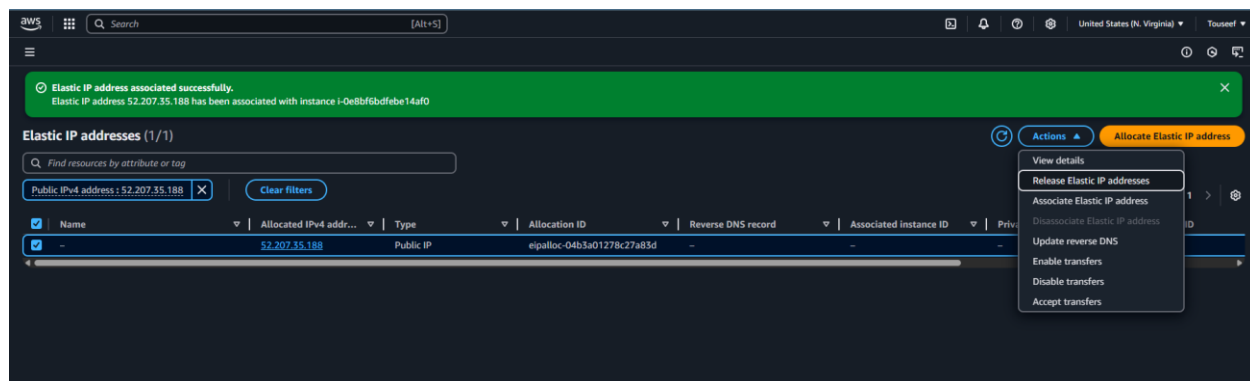
Now terminate the instance. You would notice that that Elastic IP get auto disassociated from the instance.



Now go back to the Elastic IP addresses page and you will see it is not associated with any instance.



Now release the Elastic IP by selecting the IP and going to actions > Release Elastic IP addresses



And confirm that you do want to release the IP address.

